

Statistical Modeling

DATA.STAT.740

Lecture 3

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Data transformations

Explanatory variables and outcome variables can be transformed to

- A) make the model assumptions and data values more coherent
- B) improve the interpretability of the models and model fits

Data transformations (predictors)

A) make the model assumptions and data values more coherent

The effect is proportional rather than additive If a one-unit increase in x does not have a constant effect on y , but a percentage increase does, a log transform is appropriate.

Example: going from €10 → €20 matters much more than €1,010 → €1,020.

Modeling $\log(x_1)$ captures “percent change in x that leads to additive change in y .”

$$\begin{aligned} x_1 \rightarrow x'_1 \Rightarrow \Delta y_i &= (\beta_0 + \beta_1 \log(x'_1)) - (\beta_0 + \beta_1 \log(x_1)) \\ &= \beta_1 [\log(x'_1) - \log(x_1)] = \beta_1 \log \left(\frac{x'_1}{x_1} \right) \end{aligned}$$

it's the ratio that matters

Data transformations (predictors)

One can always do an affine transformation on continuous explanatory variables

$$\begin{aligned} \mathbf{y}_i &= \beta_0 + \beta_1 x_{i,1} + \beta_1 x_{i,2} + \cdots + \beta_p x_{i,p} + \epsilon_i \\ &= \beta_0 + \beta_1 \frac{\tau_1(x_{i,1} - m_1 + m_1)}{\tau_1} + \cdots + \beta_p \frac{\tau_p(x_{i,p} - m_p + m_p)}{\tau_p} + \epsilon_i \\ &= \beta_0 + \beta_1 m_1 + \cdots + \beta_p m_p + \beta_1 \tau_1 \frac{x_{i,1} - m_1}{\tau_1} + \cdots + \beta_p \tau_p \frac{x_{i,p} - m_p}{\tau_p} + \epsilon_i \\ &= \beta'_0 + \beta'_1 x'_{i,1} + \cdots + \beta'_p x'_{i,p} + \epsilon_i. \end{aligned}$$

Data transformations

One can always do an affine transformation on continuous explanatory variables

$$\begin{aligned}y_i &= \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_p x_{i,p} + \epsilon_i \\&= \beta_0 + \beta_1 \frac{\tau_1(x_{i,1} - m_1 + m_1)}{\tau_1} + \cdots + \beta_p \frac{\tau_p(x_{i,p} - m_p + m_p)}{\tau_p} + \epsilon_i \\&= \beta_0 + \beta_1 m_1 + \cdots + \beta_p m_p + \beta_1 \tau_1 \frac{x_{i,1} - m_1}{\tau_1} + \cdots + \beta_p \tau_p \frac{x_{i,p} - m_p}{\tau_p} + \epsilon_i \\&= \beta'_0 + \beta'_1 x'_{i,1} + \cdots + \beta'_p x'_{i,p} + \epsilon_i.\end{aligned}$$

Z-transformation: $m_j = \frac{1}{N} \sum_{i=1}^N x_{i,j}$ Intercept β'_0 is the value that all: $x_{i,j} = m_j$

$\tau_j^2 = \frac{1}{N-1} \sum_{i=1}^N (x_{i,j} - m_j)^2$ Slopes β'_i : the change of y_i when x'_i changes one τ_j

Example: Data transformations

Dataset: tetrahymena.txt

Consider the research problem where there was interest to determine whether glucose had an effect on single-cell Tetrahymena cell size when cells were grown in cell culture on the growth medium. The cell size was measured by the (average) diameter of cells y_i . In research, explanatory variables were the concentration of Tetrahymena cells x_1 and glucose x_2 which was coded

$$x_2 = \begin{cases} \text{yes} & \text{Glucose was added to growth medium} \\ \text{no} & \text{Glucose was not added to growth medium} \end{cases}$$

Generalized linear models

In GLMs we allow the outcome to have some other distribution than normal

We model a transformed **mean** of the outcome with a linear combination of predictors

GLMs have three components

$\mathbf{y}_i \sim p(y \mid X)$ Random component

$\eta_i = x_i^T \beta$ Systematic component

$\eta_i = g(\mu_i)$ Link function

Random component

The distribution of the observations

$$y_i \sim p(y \mid X)$$

$$\eta_i = x_i^T \beta$$

$$\eta_i = g(\mu_i)$$

In GLMs the distributions may be of the **exponential family of distributions**

The distribution can be written as

$$f(y_i \mid \theta_i, \phi) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right) \quad (\text{bat})$$

Examples of the distributions of this form are

Normal distribution

Binomial distribution

Gamma distribution

Poisson distribution

We will look into properties of (bat) later

Systematic component

A linear combination of predictors explain the mean of the outcome distribution

$$\mathbf{y}_i \sim p(y \mid X)$$

$$\eta_i = x_i^T \beta$$

$$\eta_i = g(\mu_i)$$

Linear normal model:

$$\mathbf{y}_i \sim p(y \mid X)$$

$$\eta_i = x_i^T \beta$$

$$\eta_i = \mu_i$$

Example: normal linear model with log-link

Linear normal model: $y_i \sim \text{Normal}(\mu_i, \sigma^2)$

$$\eta_i = x_i^T \beta$$

$$\eta_i = g(\mu_i)$$

$$g(\cdot) = \log(\cdot)$$

Link functions

The explanatory variables affect the expected value through the link function

$$g(\mu_i) = x_i^T \beta$$

We have a linear combination of predictors $\eta_i = x_i^T \beta$ but μ_i is defined on a restricted scale

because e.g. Probabilities are between 0 and 1

Counts and positive outcomes are > 0

Link functions allow linear modeling of effects by transforming the original scale to a more convenient scale and their inverse transforms the new scale to original scale

Probabilities stay between 0 and 1

Counts and positive-valued responses remain positive

In practice we use link functions

$$E(\mathbf{y}_i) = x_i^T \beta \quad \text{identity link}$$

$$\log(E(\mathbf{y}_i)) = x_i^T \beta \quad \text{log link}$$

$$\log \left(\frac{E(\mathbf{y}_i)}{1 - E(\mathbf{y}_i)} \right) = x_i^T \beta \quad \text{logit link}$$

$$\frac{1}{E(\mathbf{y}_i)} = x_i^T \beta \quad \text{inverse link}$$

$$\sqrt{E(\mathbf{y}_i)} = x_i^T \beta \quad \text{square root link}$$

$$E(\mathbf{y}_i)^k = x_i^T \beta \quad \text{power link k}$$

Logistic regression (categorical observation model)

Consider the situation where the outcome y_i has only two possible values

$$y_i \in \{Y_1, Y_2\}$$

We can always code the output as $y_i \in \{0, 1\}$ (success/failure)

The binary random variable y_i follows Bernoulli distribution $y_i \mid x_i \sim \text{Ber}(\mu_i)$ where

$$P(y_i = 1 \mid x_i) = \mu_i, P(y_i = 0 \mid x_i) = 1 - \mu_i$$

Probability mass function of Bernoulli distributed random variable

$$P(y_i = y_i \mid x_i) = \mu_i^{y_i} (1 - \mu_i)^{1-y_i}, \quad y_i \in \{0, 1\}$$

Mean and variance are

$$E(y_i \mid x_i) = \mu_i, \quad \text{Var}(y_i \mid x_i) = \mu_i(1 - \mu_i)$$

Logistic regression

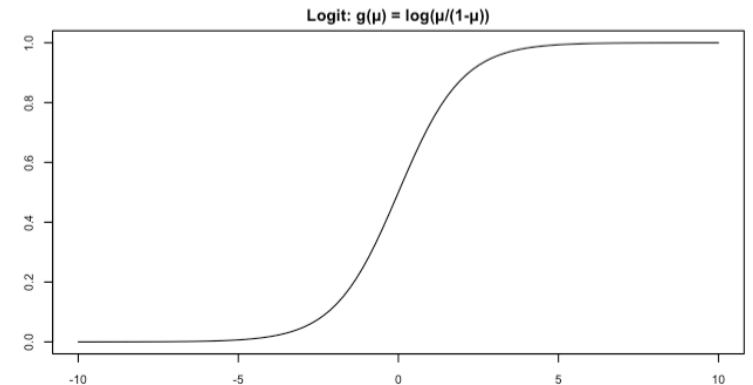
Under Bernoulli $y_i \mid x_i \sim \text{Ber}(\mu_i)$ the standard link function is the logit link

$$\text{logit}(\mu_i) = \log \left(\frac{\mu_i}{1 - \mu_i} \right) = x_i^T \beta$$

$\uparrow$
odds = $\frac{P(y_i = 1)}{P(y_i = 0)}$

The inverse link induces the expected value to have a form

$$\mu_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}$$



Logistic regression

$$\text{logit}(\mu_i) = \log \left(\frac{\mu_i}{1 - \mu_i} \right) = x_i^T \beta$$

We usually quantify the effect of the predictors using odds ratios

Let

$$\log \left(\frac{\mu_i}{1 - \mu_i} \right) = \beta_0 + \beta_1 x_{i,1} + \cdots \beta_p x_{i,p}$$

$$\Leftrightarrow \frac{\mu_i}{1 - \mu_i^{(1)}} = \exp(\beta_0 + \beta_1 x_{i,1} + \cdots \beta_p x_{i,p}) =: \text{Odds}(x_1)$$

and

$$\text{Odds}(x_1 + 1) = \exp(\beta_0 + \beta_1(x_{i,1} + 1) + \cdots \beta_p x_{i,p}) = \exp(\beta_0 + \beta_1 x_{i,1} + \beta_1 + \cdots \beta_p x_{i,p})$$

Odds ratio for $x_{i,1}$ is defined for a one unit increase in $x_{i,1}$

$$\frac{\text{Odds}(x_1 + 1)}{\text{Odds}(x_1)} = \exp(\beta_1)$$

Logistic regression

Odds ratio for $x_{i,1}$

$$\frac{\text{Odds}(x_1 + 1)}{\text{Odds}(x_1)} = \exp(\beta_1)$$

Interpretation

$\beta_1 = 0 \Rightarrow \exp(\beta_1) = 1$: No effect

$\beta_1 > 0 \Rightarrow \exp(\beta_1) > 1$: The odds increase

$\beta_1 < 0 \Rightarrow \exp(\beta_1) < 1$: The odds decrease

Example: Logistic regression

Dataset: malaria.txt

A random sample of 100 children aged 3-15 (x_1) years from a village in Ghana. The children were followed for a period of 8 months. At the beginning of the study, values x_2 of a particular antibody were assessed. Based on observations during the study period, the children were categorized into two groups: individuals with and without symptoms of malaria y_i .

Poisson regression (count data model)

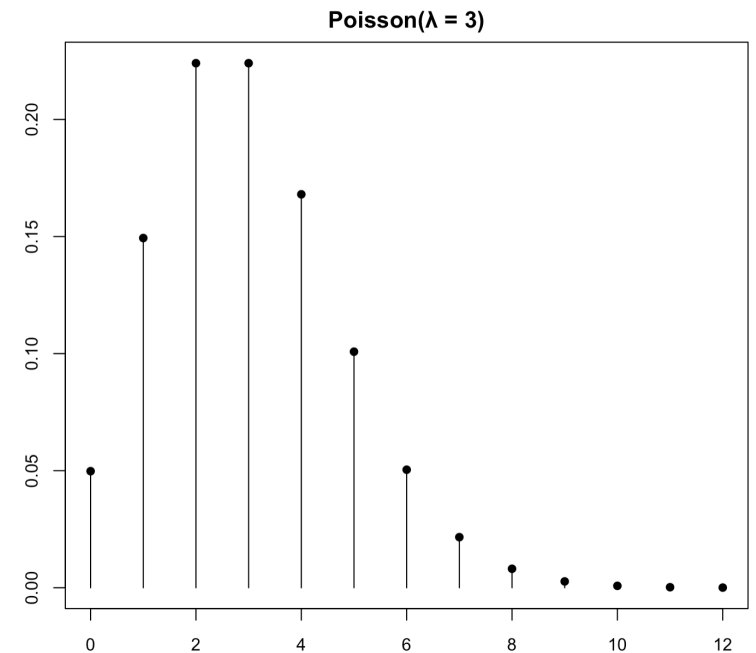
In count data models, the random response y_i can have a realization as a nonnegative integer $y_i \in \{0, 1, 2, 3, 4, \dots\}$

Probability mass function of the Poisson distribution is

$$P(\mathbf{y}_i = k \mid x_i) = \frac{\mu_i^k \exp(-\mu_i)}{k!}$$

Mean and the variance are

$$E(\mathbf{y}_i \mid x_i) = \text{Var}(\mathbf{y}_i \mid x_i) = \mu_i$$



Poisson regression

In Poisson regression we most often use log-link $\log(\mu_i) = x_i^T \beta$

Because $\mu_i = \exp(x_i^T \beta)$, a one unit increase in $x_{i,1}$ affects the mean

$$\mu_i^{(1)} = \exp(x_i^T \beta)$$

$$\mu_i^{(2)} = \exp(\beta_0 + (x_{i,1} + 1)\beta_1 + \cdots + x_{i,p}\beta_p) = \exp(\beta_1)\mu_i^{(1)}$$

$\exp(\beta_1)$: rate ratio/incidence rate ratio

Example: Poisson regression

Dataset: species.txt

The response is a count of the number of plant species on plots that have different biomass (a continuous explanatory variable x_1) and different soil pH (a categorical variable x_2 with three levels: high=1, mid=2, and low=3).

Poisson regression with offset

Poisson regression is a model for **counts**

For example count could be the number of events happening within a time interval

What if the time intervals of data collection are **not** the same size?

You would expect more counts from larger intervals

Let's formulate the model using ratio of counts per index variable instead

Expected value of the ratio is

$$E\left(\frac{y_i}{t_i}\right) = \frac{\mu_i}{t_i}$$

Poisson regression with offset

log link function is the most suitable model for the expected value of the ratio

$$\log \left(\frac{\mu_i}{t_i} \right) = x_i^T \beta$$

We can reformulate the ratio model as

$$\log(\mu_i) = \log(t_i) + x_i^T \beta$$


offset

We're back on Poisson regression model with added offset

End of Lecture 3